

Antibacterial Screening Compound Libraries

Infectious diseases are one of the most critical and urgent health problems in the world. Although the market for antibacterial drugs is nowadays greater than 25 billion US dollars per year, the development and implementation of novel antibacterial medicines are still required due to the developed resistance of many pathogenic bacteria against current antibiotics [1].

Since there are typically several antibacterial agents whose effectiveness against many pathogens is comparable, they can be used empirically and the selection of the treatment approach often depends on various indirect factors, such as pharmacokinetics, side effects, resistance profile, and treatment costs [2]. That is why the development of novel antibacterial drugs that focus on target-activity relationships and belong to structurally different chemical classes still remains a pressing matter (Fig. 1) [3].

Life Chemicals has developed dedicated **Antibacterial Screening Compound Libraries** of over **18,000** drug-like screening compounds with potential bacterial inhibitory activity as excellent starting points for high throughput screening (HTS) and high content screening (HCS) projects, as well as hit identification in antibacterial drug discovery:

- [Antibacterial Focused Library](#)
- [Antibacterial Targeted Library](#)

These two non-overlapping Screening Sets were carefully selected with both ligand-based (2D fingerprint similarity search) and structure-based (virtual screening, molecular docking) approaches.



The compound selection can be customized based on your requirements, cherry-picking is available.



Please, contact us at orders@lifechemicals.com for any additional information and price quotations.



In their pre-plated format the Sets can be found within our exceptional collection of **Pre-plated Focused Libraries**.



You can also be interested in the following related products:

- [Antituberculosis Screening Compound Libraries](#)
- [Covalent Screening Library](#)
- [Antiviral Screening Compound Libraries](#)

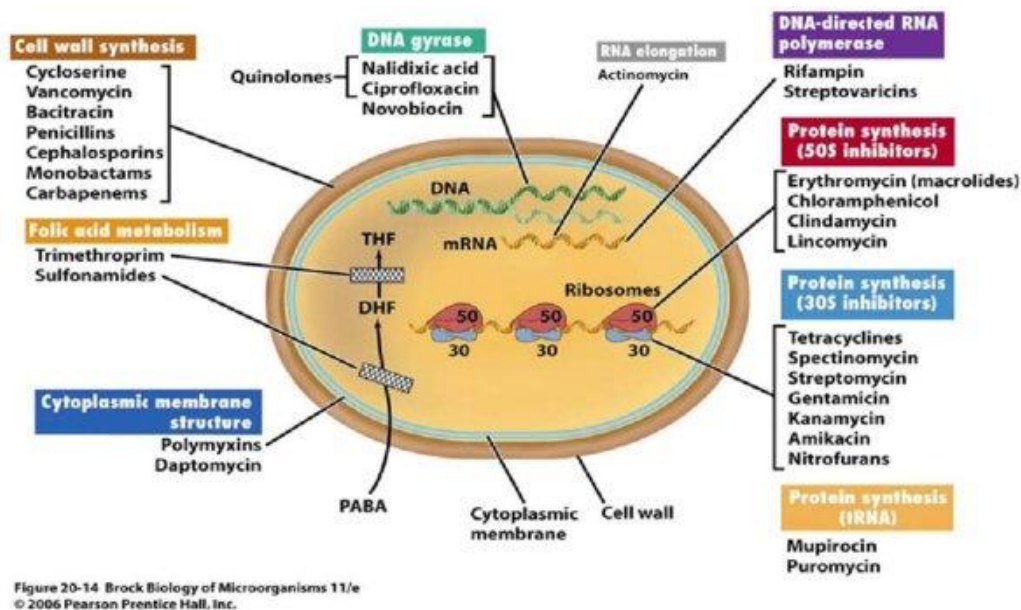


Figure 1. The key cell process and protein targets for commonly used antimicrobial agents. Picture credit: Yazdankhah, S et al., 2018 [4].

Antibacterial Focused Library

A 2D fingerprint similarity search (Tanimoto similarity index > 0.80, < 10 analogs per reference to ensure diversity) of the Life Chemicals **HTS Compound Collection** allowed us to select around **15,500** drug-like screening compounds with potential antibacterial activity.

A reference set of compounds with reported antibacterial activity against different gram-positive and gram-negative bacteria species (provided below) was prepared. Activity data threshold was chosen < 10 μ M, extracted from Binding and ChEMBL databases. As a result, we picked around **8,900** screening compounds effective against bacterial organisms and over **6,900** small-molecule analogues of compounds with known activity against the following bacteria-related protein targets:

Streptokinase A	Carbonic anhydrase	Transcriptional regulator MvfR
ATP-dependent Clp protease	Bifunctional protein GlmU	Beta-galactosidase
4'-phosphopantetheinyl transferase	Quinolone resistance protein norA	alpha/beta hydrolase fold family
Protein RecA	DNA topoisomerase	mRNA interferase MazF
Anthrax lethal factor	Chorismate synthase	Dihydrofolate reductase
Fructose-bisphosphate aldolase	UDP-N-acetylmuramoyl-tripeptide--	Signal transduction protein TRAP
Aminotransferase	D-alanyl-D-alanine ligase	15-cis-phytoene desaturase
Beta-lactamase (type AmpC, NDM-1, TEM, VIM-2)	Pyruvate kinase	Autoinducer 1 sensor
Enoyl-[acyl-carrier-protein] reductase	Botulinum neurotoxin (type A, E, F)	kinase/phosphatase luxN
Replicative DNA helicase	Pantothenate synthetase	Histidine biosynthesis bifunctional protein HisB
DNA gyrase (subunit A, B)	Thymidylate synthase	Tyrosine-protein phosphatase yopH
UDP-D-alanine ligase	UDP-galactopyranose mutase	3-oxoacyl-[acyl-carrier-protein] synthase 3
Dihydrolipoyl dehydrogenase	Peptide deformylase	Cereblon isoform 4
Probable nicotinate-nucleotide adenyltransferase	Taq polymerase 1	Epoxide hydrolase
Shiga toxin	Lanosterol 14-alpha demethylase	GroEL/GroES
Probable L-lysine-epsilon aminotransferase	Acyl-CoA synthase	UDP-3-O-[3-hydroxymyristoyl] N-acetylglucosamine deacetylase
HTH-type transcriptional regulator	Carbonate dehydratase	Dihydrodipicolinate synthase
Exoenzyme S	Inosine-5'-monophosphate dehydrogenase	Lectin
Histidine protein kinase DivJ	Transcriptional activator (type lasR, luxR, traR)	rRNA adenine N-6-methyltransferase
	Tyrosine-protein phosphatase PTPB	

Accessory gene regulator protein A
Dipeptidyl peptidase IV
Heme oxygenase
Methionine aminopeptidase
Uncharacterized protein
Rv1284/MT1322
2-heptyl-4(1H)-quinolone synthase
PqsD
Chaperone protein dnaK
Cytochrome P450
Dehydrosqualene desaturase
Dihydropteroate synthase
Diphosphomevalonate decarboxylase

Glutathione-independent
formaldehyde dehydrogenase
Lycopene cyclase
Phenylalanyl-tRNA synthetase
Pseudolysin
Squalene-hopene cyclase
Thermolysin
UDP-N-acetylglucosamine N-
acetyltransferase
Urease (subunit alpha, beta)
Virulence sensor histidine kinase 5-
enolpyruvylshikimate-3-phosphate
synthase

CAI-1 autoinducer sensor
kinase/phosphatase CqsS
CpG DNA methylase
Cystathionine beta-lyase metC
D-alanyl-D-alanine carboxypeptidase
Dehydrosqualene synthase
DNA polymerase III
Histidinol dehydrogenase
Prolyl endopeptidase
Protoporphyrinogen oxidase
Thymidylate kinase
Toxin B
tRNA-guanine transglycosylase

Bacterial species cited in the reference compound set:

Acinetobacter baumannii
Actinomyces viscosus
Aeromonas hydrophila
Agrobacterium tumefaciens
Alcaligenes faecalis
Alicyclobacillus acidocaldarius
Allivibrio fischeri
Aquifex aeolicus
Bacillus (*B.anthraxis*, *B.cereus*,
B.megaterium, *B.subtilis*,
B.thermoproteolyticus)
Bordetella bronchiseptica
Brucella suis
Burkholderia cenocepacia
Campylobacter jejuni
Caulobacter vibrioides
Chlamydomonas pneumoniae
Citrobacter freundii
Clavibacter michiganensis
Clostridium botulinum
Corynebacterium glutamicum

Elizabethkingia meningoseptica
Enterococcus faecalis
Erwinia chrysanthemi
Escherichia coli
Flavobacterium columnare
Francisella tularensis
Haemophilus influenzae
Helicobacter pylori
Klebsiella pneumoniae
Lactobacillus casei
Listeria innocua
Listonella anguillarum
Magnetospirillum gryphiswaldense
Micrococcus luteus
Moraxella catarrhalis
Mycobacterium (*M.bovis* wildlife,
M.bovis BCG, *M.chelonae*, *M.fortuitum*,
M.kansasii, *M.peregrinum*,
M.smegmatis, *M.tuberculosis*)
Mycoplasma gallisepticum
Pantoea ananas
Peptoclostridium difficile

Porphyromonas gingivalis
Propionibacterium acnes
Proteus vulgaris
Pseudomonas (*P.aeruginosa*, *P.putida*)
Raoultella planticola
Salmonella (*S.choleraesuis*, *S.enterica*,
S.typhimurium)
Shigella (*S.dysenteriae*, *S.flexneri*)
Spiroplasma monobiae
Staphylococcus (*S.aureus*,
S.epidermidis)
Streptococcus (*S.agalactiae*,
S.mutans, *S.pneumoniae*, *S.pyogenes*)
Sulfurihydrogenibium
Synechococcus elongatus
Thermus aquaticus
Thiomicrospira crunogena
Vibrio (*V.cholerae*, *V.harveyi*)
Yersinia (*Y.enterocolitica*, *Y.pestis*,
Y.pseudotuberculosis)
Zymomonas mobilis

Additionally, approved antibacterial compounds for systemic use and their analogs were also included:

Azithromycin
Ceftazidime
Ceftazoxime
Cinoxacin
Ciprofloxacin
Flumequine
Linezolid
Mandelic acid
Metronidazole
Nalidixic acid

Nitrofurantoin
Oxolinic acid
Pefloxacin
Procaine benzylpenicillin
Rosoxacin
Roxithromycin
Sulfadimethoxine
Sulfamerazine
Sulfamethazine
Sulfamethizole

Sulfamethoxazole
Sulfametomidine
Sulfanilamide
Sulfapyridine
Sulfathiazole
Sulfathiourea
Sulfisomidine
Sulfisoxazole
Trimethoprim

Antibacterial Targeted Library

Due to the persistent spread of the epidemic of coronavirus infection, more and more attention is being paid to the research of the causative agents of respiratory tract diseases and the development of effective medicines against them.

Neisseria meningitidis and ***Streptococcus pneumoniae*** represent the causative agents of invasive bacterial infections, manifested by a wide range of diseases, such as meningitis and septicemia [5–7]. ***Klebsiella pneumoniae*** is the causative agent of both pneumonia and

nosocomial human infections, characterized by high virulence and resistance to antibiotics, especially carbapenem class [8, 9]. These pathogens are commonly spread worldwide and can lead to fatal consequences in case of a severe course of the disease.

Addressing these urgent problems, our cheminformatics team has designed this Screening Set of around **2,700** structurally-diverse screening compounds with potential biological activity. The molecules were picked out by virtual molecular screening against the most relevant pneumonia-causing bacterial drug targets important for viability, virulence, and resistance of the above-mentioned microorganisms:

- [N-acetylneuraminate synthase \(NeuB\)](#)
- [Metallo- \$\beta\$ -lactamase \(MBL\)](#)
- [NanB](#)

N-acetylneuraminate synthase (NeuB)

N-acetylneuraminate synthase (NeuB) is a bacterial sialic acid synthase, vital for neuroinvasive bacteria, such as *Neisseria meningitidis*, enabling them to synthesize N-acetylneuraminate (NeuNAc) to evade the host immune system. This makes NeuB a highly relevant drug target for antibacterial drug development [10].

Key features:

- **Method:** high-throughput virtual screening (pharmacophore hypothesis)
- **X-Ray data used:** 6PPY
- **Filters used:** QikProp properties and descriptors
- **Number of compounds selected:** 586

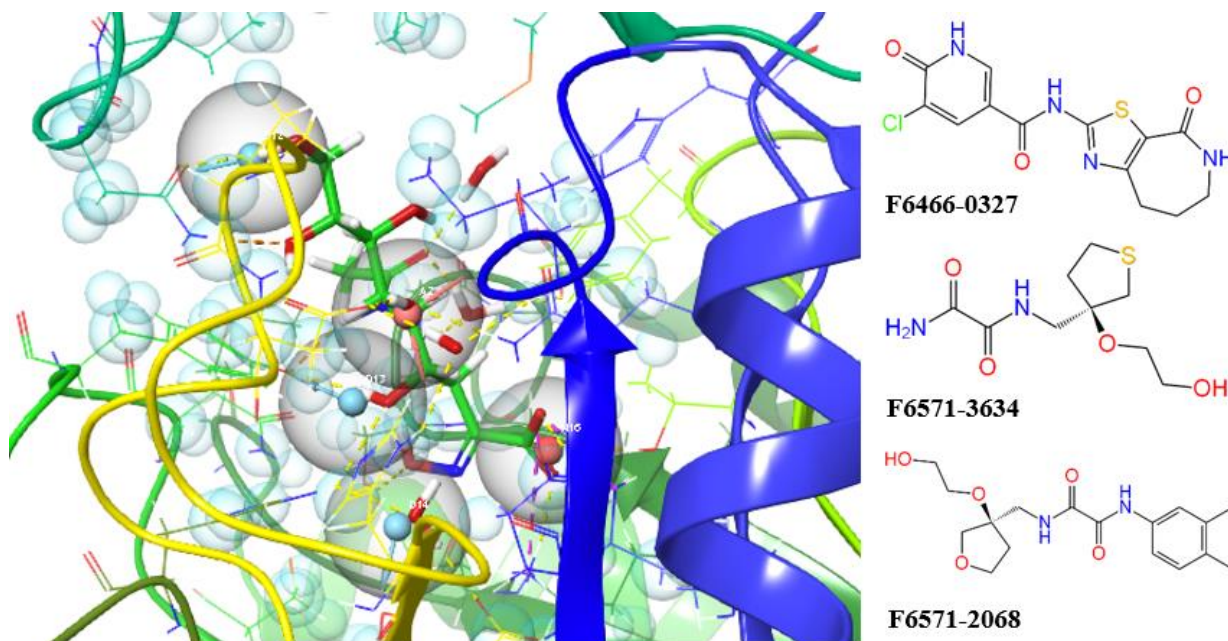


Figure 2. The pharmacophore hypothesis by 6PPY and examples of lead compounds.

Metallo- β -lactamase (MBL)

Antimicrobial therapy is threatened by the global rise of resistance, especially in Gram-negative bacteria, where resistance to β -lactams is largely mediated by β -lactamases. Carbapenems, which evade most β -lactamases, are hydrolyzed by metallo- β -lactamases (MBLs), as well as by a few active-

site serine β -lactamases (SBLs), that contributes to the resistance against them of such enterobacteria as *Klebsiella pneumoniae*. Despite a significant progress in the development of beta-lactamase inhibitors, effective Metallo- β -lactamase inhibitors still remain a challenge [11].

Key features:

- **Method:** high-throughput virtual screening (pharmacophore hypothesis)
- **X-Ray data used:** 6V1M
- **Filters used:** QikProp properties and descriptors
- **Number of compounds selected:** 1673

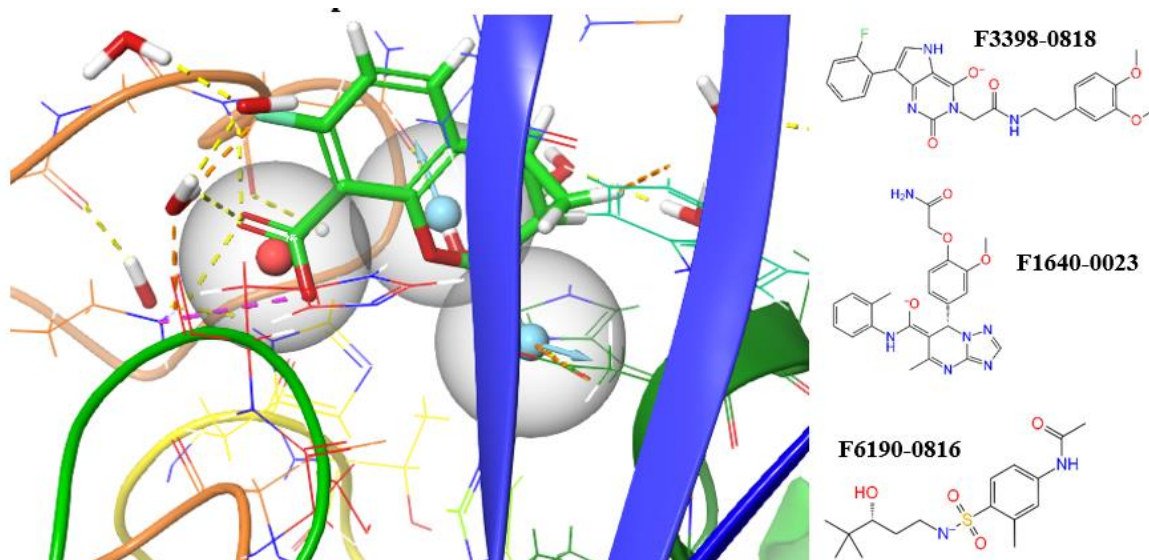


Figure 3. The pharmacophore hypothesis by 6V1M and examples of lead compounds.

NanB

Neuraminidases (sialidases) are important in the pathogenicity of bacteria and the virulence of influenza. They are key enzymes in obtaining nutrients by *Streptococcus pneumoniae*, as they are able to break down sugars on the surface of host cells, thus causing respiratory tract damage. The above makes NanB a potential drug target to inhibit bacterial infections [12].

Key features:

- **Method:** high-throughput virtual screening (pharmacophore hypothesis)
- **X-Ray data used:** 4XHB
- **Filters used:** QikProp properties and descriptors
- **Number of compounds selected:** 855

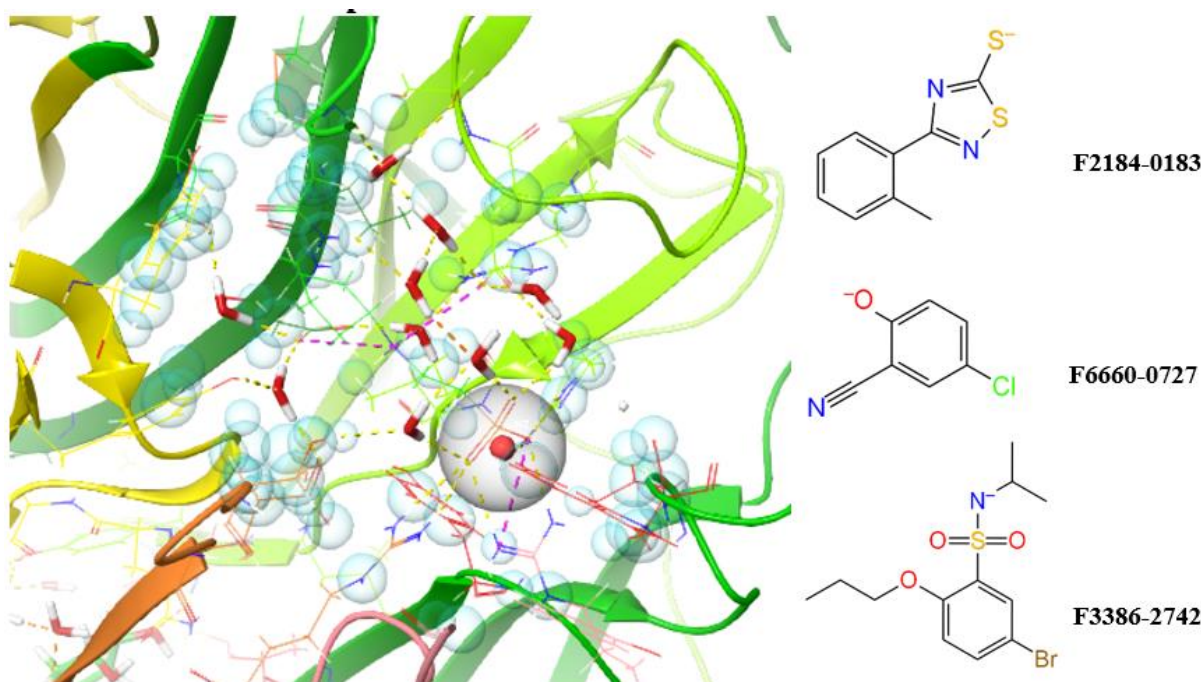


Figure 4. The pharmacophore hypothesis by 4XHB and examples of lead compounds.

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